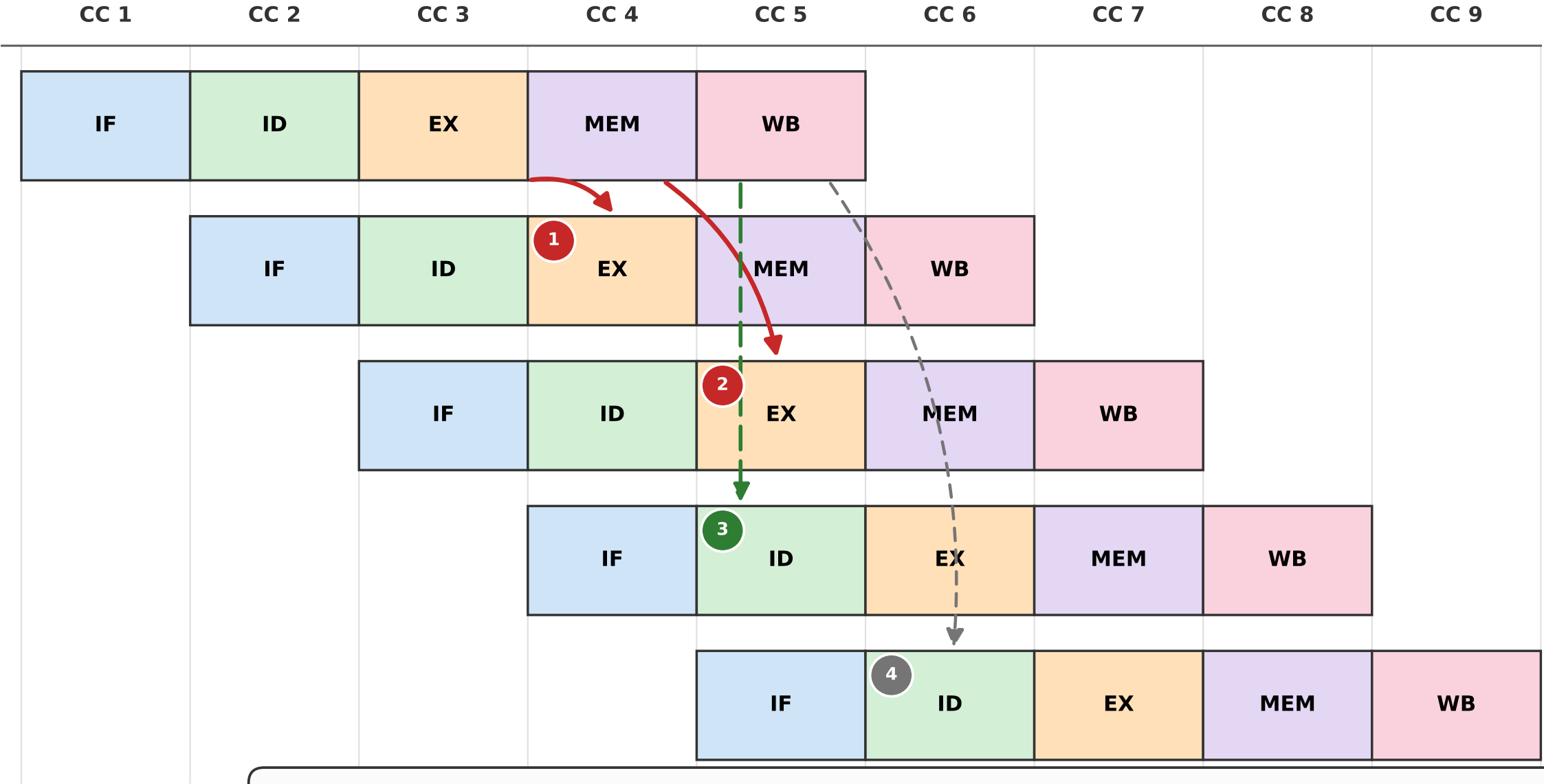


Q4 — Pipeline Time Diagram: Data Dependencies & Forwarding

sub \$2,\$1,\$3 → and \$12,\$2,\$5 → or \$13,\$6,\$2 → add \$14,\$2,\$2 → sw \$15,100(\$2)

RAW dependency
on \$2



- Forwarding path REQUIRED
- Solved by register file (same cycle)
- No hazard
- Data dependency (RAW)

sub produces \$2 at the END of EX (CC3) but writes the register file only in WB (CC5).

- 1 and needs \$2 in EX (CC4) → FORWARD from the EX/MEM pipeline register.
- 2 or needs \$2 in EX (CC5) → FORWARD from the MEM/WB pipeline register.
- 3 add reads \$2 in ID (CC5) while sub writes it in WB (CC5) → register file writes in the 1st half of the cycle and reads in the 2nd half, so NO forwarding is needed.
- 4 sw reads \$2 in ID (CC6), after WB has finished → NO hazard at all.

RESULT: 2 forwarding paths, 0 stalls, total = 5 + (5 – 1) = 9 clock cycles.